TITAN : A DECENTRALIZED THREAT INTELLIGENCE SHARING USING BLOCKCHAIN

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ABSTRACT

Cyberattacks are rapidly increasing, making it essential for organizations to share Cyber Threat Intelligence (CTI) to improve overall security. However, most existing CTI sharing systems are centralized and depend heavily on manual trust between organizations. This often results in the spread of fake or low-quality threat data and creates hesitation among organizations to share sensitive information. To address these challenges, the proposed system introduces a decentralized, blockchain-based CTI sharing model. Blockchain technology ensures transparency, immutability, and tamper-evident records for all shared intelligence. In addition, a reputation scoring mechanism is used to evaluate the reliability of participants and encourage honest contributions. This approach improves trust, data quality, and collaboration among organizations, enabling a more secure and efficient cyber threat intelligence sharing environment.

INTRODUCTION

- Cyberattacks are increasing rapidly, creating the need for organizations to share Cyber Threat Intelligence (CTI).
- Existing CTI sharing systems are centralized, leading to trust issues, data manipulation, and a single point of failure.
- Lack of transparency and proper verification reduces confidence among participating organizations.
- TITAN proposes a decentralized, blockchain-based CTI sharing system with reputation scoring to ensure trust and reliability.
- Blockchain technology provides tamper-proof and transparent records of all shared threat intelligence.
- The reputation scoring mechanism helps identify trustworthy contributors and filter low-quality data.
- Decentralization removes dependency on a single authority, improving system security and availability.
- The system encourages secure collaboration and faster sharing of threat information among organizations.
- This approach improves data integrity, trust, and efficiency in cyber threat intelligence sharing.

EXISTING SYSTEM & DISADVANTAGES

Existing System:

- **Information sharing is coordinated through a central server, and participant trust is informal, manual, and slow to build.**

Disadvantages:

- **Single Point of Failure (SPOF): Central servers are highly vulnerable to manipulation and targeted attacks.**
- **Freeriding: Many organizations consume valuable threat data without contributing anything back to the network.**
- **Low Data Quality: Without automated verification, fake or outdated threat data easily circulates.**
- **Lack of Auditability: Trust cannot be mathematically proven or tracked over time.**

LITERATURE REVIEW (PAPER 1)

- **Title of Paper:** [Y. Wu, Y. Qiao, Y. Ye, B. Lee] Towards Improved Trust in Threat Intelligence Sharing Using Blockchain and Trusted Computing
- **Objective:** Aims to increase trust and safety among users when sharing cybersecurity threat data.
- **Methodology:** Combines a blockchain network, secure computing, and user reputation scores to verify shared data.
- **Key Findings:** Shows that blockchain makes shared data transparent, highly reliable, and safe from being secretly altered.
- **Limitations:** Focuses mostly on theoretical trust models rather than testing how the system handles large-scale, real-world use.
- **Relevance:** Acts as the foundational blueprint for your project's (TITAN's) own threat-sharing and reputation system.

LITERATURE REVIEW (PAPER 2)

- **Title:** (S. Li, T. Jiang, K. Chen) Blockchain-based Secure Information Sharing Framework for Cyber Threat Intelligence
- **Objective:** To develop a secure decentralized framework for CTI sharing using blockchain.
- **Methodology:** Utilizes smart contracts for CTI validation, a distributed ledger for storing hash values, and off-chain storage for large threat data.
- **Key Findings:** Eliminates the need for a central authority, ensures tamper-proof data records, and significantly improves data integrity.
- **Limitations:** Faces performance issues in large-scale networks, high gas costs on the Ethereum network, and delays due to blockchain transaction times.
- **Relevance:** Directly supports TITAN's technical architecture by validating the use of Ethereum and smart contracts for secure data sharing.

LITERATURE REVIEW (PAPER 3)

- **Title of Paper:** (X. Zhang, L. Liu) A Reputation-based Trust Management Model for Information Sharing.
- **Objective**: To design a robust trust evaluation mechanism for secure and reliable information exchange.
- **Methodology**: Implements a reputation scoring system using trust aggregation algorithms and a peer-to-peer evaluation model.
- **Key Findings**: Demonstrates that reputation systems improve honest participation and effectively detect malicious contributors within the network.
- **Limitations**: The system remains vulnerable to collusion attacks and trust manipulation, requiring high computational complexity for global trust calculations.
- **Relevance**: Directly inspires the core logic of TITAN's EigenTrust-based reputation scoring to ensure data quality.

LITERATURE REVIEW (PAPER 4)

- **Title of Paper:** (M. Patel, R. Shah) Decentralized Cyber Threat Intelligence Sharing Using Smart Contracts.
- **Objective**: Aims to eliminate single points of failure in CTI platforms by utilizing decentralized blockchain architecture.
- **Methodology**: Employs smart contracts for participant registration and rating, a transparent blockchain ledger, and automated trust evaluation.
- **Key Findings**: Effectively prevents "freeriding," encourages active contribution from all participants, and enhances the overall trust of the ecosystem.
- **Limitations**: Faces scalability concerns, network congestion risks, and high energy consumption typical of standard blockchain implementations.
- **Relevance**: Directly relates to the core decentralized framework of TITAN by validating the use of automated trust and smart contracts.

LITERATURE REVIEW (PAPER 5)

- **Title of Paper:** (S. Kamvar, M. Schlosser, Garcia-Molina) EigenTrust Algorithm for Reputation Management in Distributed Systems.
- **Objective**: Designed to compute reliable global trust scores in decentralized, peer-to-peer systems to minimize the impact of malicious nodes.
- **Methodology**: Employs an Eigenvector-based trust calculation combined with iterative reputation aggregation across the network.
- **Key Findings**: Proves highly effective at accurately identifying malicious nodes and providing a verifiable global trust ranking.
- **Limitations**: The model is sensitive to the initial trust values assigned and can be computationally intensive as the number of nodes increases.
- **Relevance**: Serves as the mathematical foundation for TITAN’s reputation system, specifically for its global trust computation logic.

LITERATURE REVIEW (PAPER 6)

- **Title of Paper:** (A. Dorri, S. Kanhere, R. Jurdak) Blockchain Technology for Secure Data Sharing in Cybersecurity.
- **Objective**: To analyze and demonstrate how blockchain architectures can fundamentally improve the security and reliability of data sharing.
- **Methodology**: Explores the implementation of a distributed ledger architecture using cryptographic hashing and various consensus mechanisms.
- **Key Findings**: Proves that blockchain ensures data immutability, successfully eliminates central authority risks, and significantly enhances network transparency.
- **Limitations**: Highlights high energy consumption and scalability issues as primary barriers, alongside the need for costly infrastructure.
- **Relevance**: Provides the essential theoretical background and architectural validation for TITAN's core blockchain design.

COMPARATIVE ANALYSIS

System / Paper	Trust Mechanism	Storage Strategy	Key Limitations
Traditional CTI	Manual / Informal	Centralized Server	Single Point of Failure, Freeriding
Wu et al., 2019	Trusted Computing	Blockchain ledger	High computational overhead
Li et al.	Smart Contracts	On-chain hash / Off-chain data	Ethereum gas costs
TITAN (Proposed)	EigenTrust Algorithm	IPFS/Off-chain + Polygon/Eth Smart Contracts	Network scalability challenges

RESEARCH GAPS

- **Reliance on Theoretical Trust**: Most current frameworks lack empirical testing for real-time, large-scale cybersecurity threat data.
- **Vulnerability to Collusion**: Existing reputation models struggle to prevent "Sybil" attacks where malicious nodes artificially inflate each other's trust scores.
- **High On-Chain Latency & Costs**: Executing threat validation directly on-chain leads to prohibitive gas fees and slow transaction speeds.
- **The "Freerider" Incentive Gap**: A lack of robust mechanisms to balance the cost of participation with high-quality, verified data contributions.
- **Computational Complexity**: Robust algorithms like EigenTrust are often too mathematically intensive to scale efficiently in decentralized networks.
- **Metadata Centralization**: Many "decentralized" systems still rely on central databases for indexing threat data, creating a single point of failure.

PROBLEM DEFINITION & OBJECTIVES

Problem Definition:

- Single Point of Failure: Centralized CTI platforms are vulnerable to DDoS attacks and data manipulation.
- Lack of Trust: Manual verification is slow and open to collusion/sybil attacks by malicious contributors.
- Inefficiency: High transaction costs and latency prevent real-time sharing of critical threat data

Objectives:

- Build a fully decentralized framework for sharing threat intelligence without central authorities.
- Implement an automated EigenTrust model to objectively score and verify user reputation.
- Use smart contracts to ensure transparent, tamper-proof data records.
- Minimize costs and latency through optimized trust aggregation and efficient blockchain integration.
- Incentivize high-quality contributions while eliminating freeriding within the ecosystem.

ADVANTAGES & APPLICATIONS

Advantages:

- **Decentralized Trust: Removes central points of failure using a P2P blockchain network.**
- **Automated Reputation: Uses EigenTrust to filter data quality and block malicious nodes.**
- **Data Immutability: Anchors threat metadata on-chain to prevent unauthorized tampering.**
- **High Availability: Distributed storage ensures data is always accessible.**

Applications:

- **Enterprise Collaboration: Secure CTI sharing between firms without exposing secrets.**
- **Government Networks: Tamper-proof intelligence exchange for national security.**
- **Community Defense: A transparent platform for rewarding independent researchers.**

LIMITATIONS

- **Network Latency**: Block confirmation times may cause slight delays in real-time threat sharing.
- **Storage Costs**: High on-chain costs limit the storage of large malware samples directly on the ledger.
- **Initial Trust Bias**: System reliability depends heavily on the integrity of the initial "pre-trusted" nodes.
- **Scalability Overhead**: Iterative EigenTrust calculations become more resource-intensive as the user base grows.
- **Gas Fee Fluctuations**: Variable transaction costs on the Ethereum network may discourage frequent small updates.
- **Data Noise**: The system cannot entirely filter out unintentional or poorly formatted "noisy" data from legitimate users.

TECHNOLOGY STACK

Smart Contracts & Web3:

- **Solidity**
- **Ethereum Blockchain**
- **Remix IDE**
- **MetaMask**

Programming Languages:

- **Python / JavaScript**

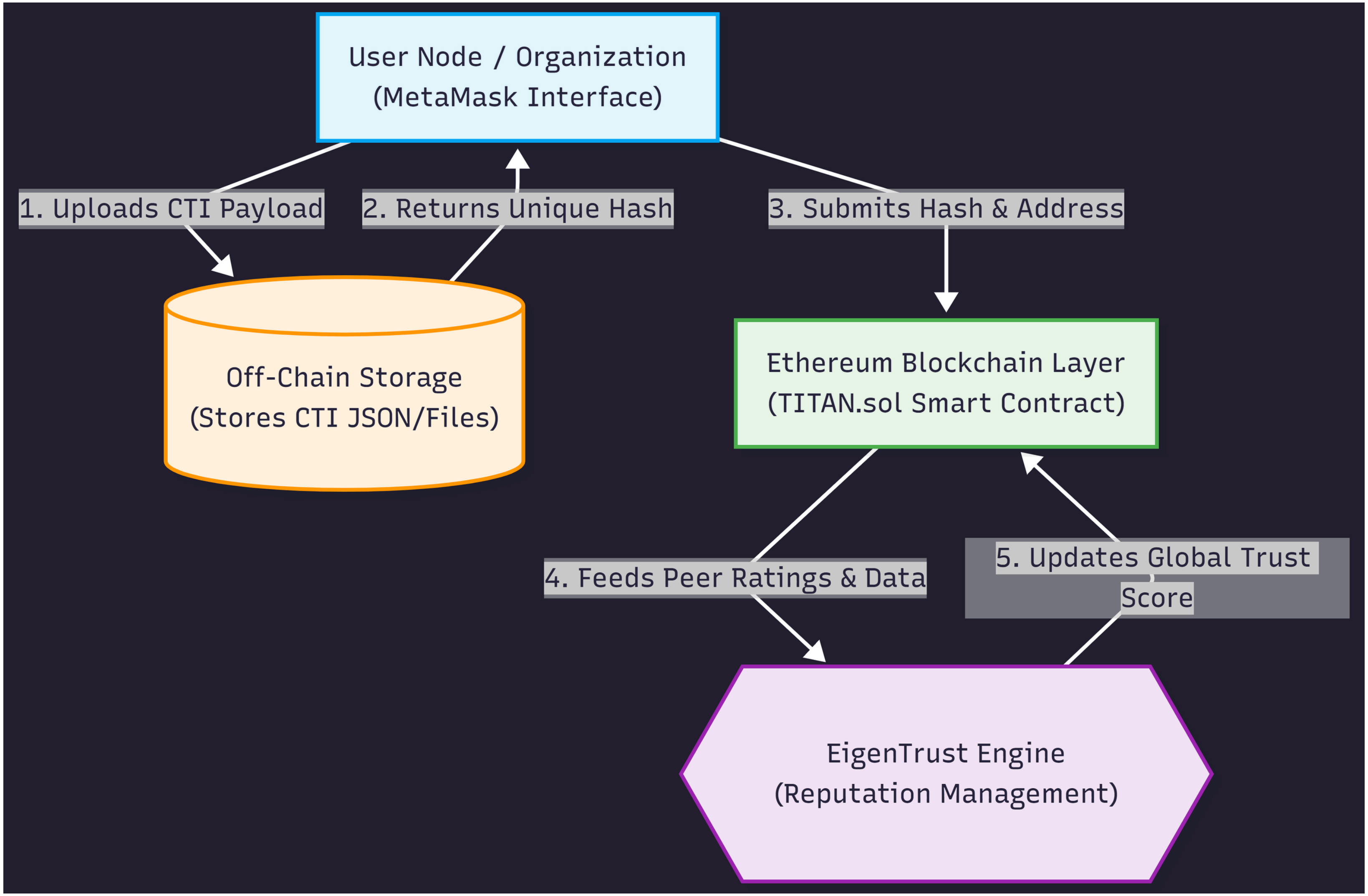
Hardware Requirements:

- **Intel i3 or above, 16 GB RAM, 256 GB HDD/SSD**
- **Stable internet.**

Operating System:

- **Windows / Linux.**

SYSTEM ARCHITECTURE



MODULE DESCRIPTION

Module 1: Authentication & Registration:

Handles wallet connection (MetaMask) and initializes a node's starting trust score on the blockchain.

Module 2: Off-Chain Storage & Hash Generation:

Captures the CTI payload, stores it off-chain to save gas, and computes a SHA-256 hash representing the data.

Module 3: Smart Contract Registry:

Logs the cryptographic hash and the submitter's Web3 address permanently onto the distributed ledger.

Module 4: Reputation Management (EigenTrust):

Aggregates peer ratings on shared intelligence to recursively update and broadcast a node's global trust ranking.

METHODOLOGY / WORKFLOW

Phase 1: Submission

- **Step 1: An organization detects a threat and compiles a CTI report.**
- **Step 2: The user connects their Web3 wallet and uploads the report.**
- **Step 3: The system saves the report off-chain and generates a unique cryptographic hash.**
- **Step 4: A smart contract transaction is triggered to permanently anchor the hash and submitter details on the blockchain.**

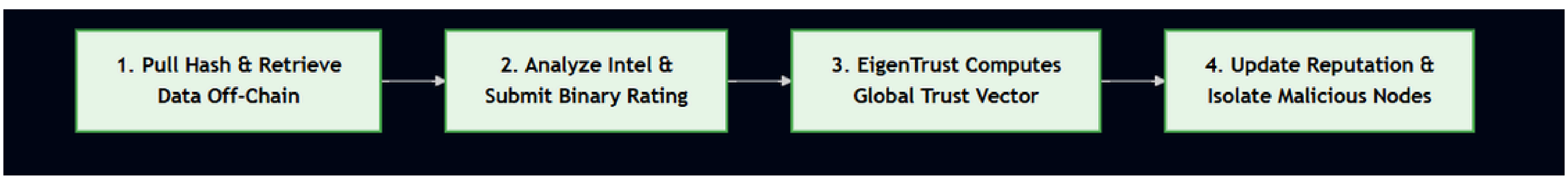
Phase 2: Validation & Trust

- **Step 1: Peer organizations pull the CTI hash from the blockchain and retrieve the data from off-chain storage.**
- **Step 2: Peers analyze the intelligence and submit a binary rating (+1 for accurate, -1 for false/malicious) back to the smart contract.**
- **Step 3: The EigenTrust algorithm computes the global trust vector.**
- **Step 4: The submitting node's reputation is dynamically updated; malicious nodes are subsequently isolated from the network.**

WORKFLOW BLOCK DIAGRAM

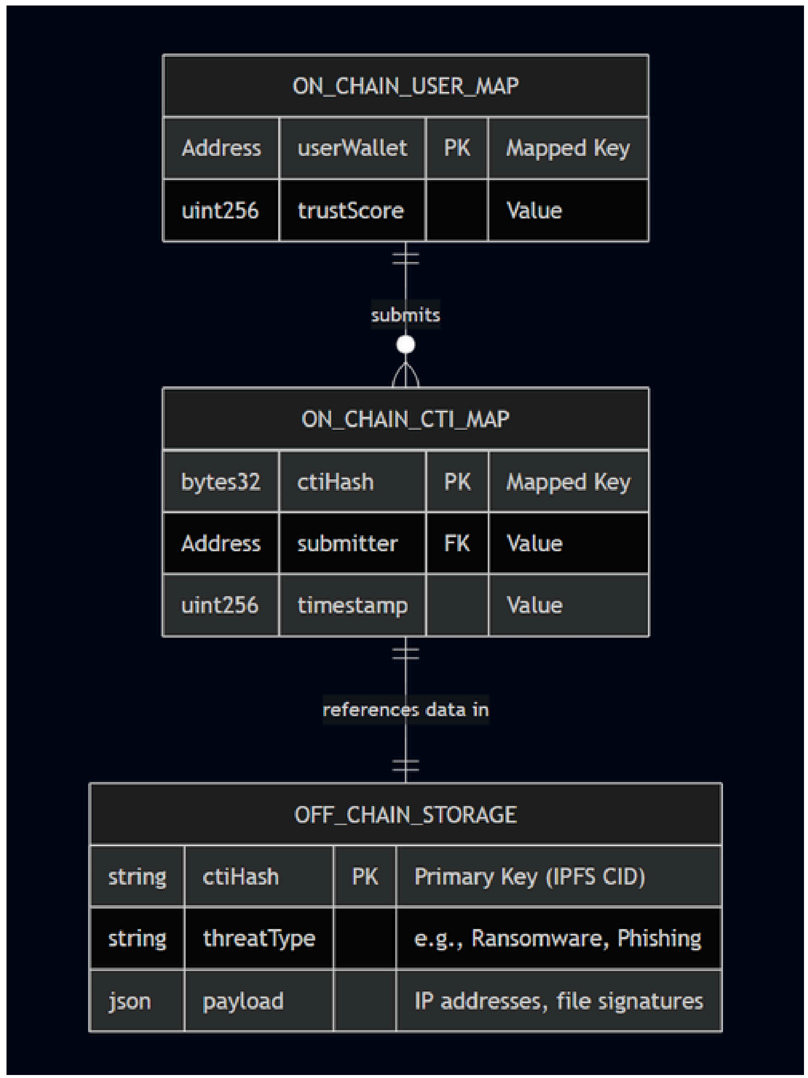
Phase 1: Submission

Phase 2: Validation & Trust



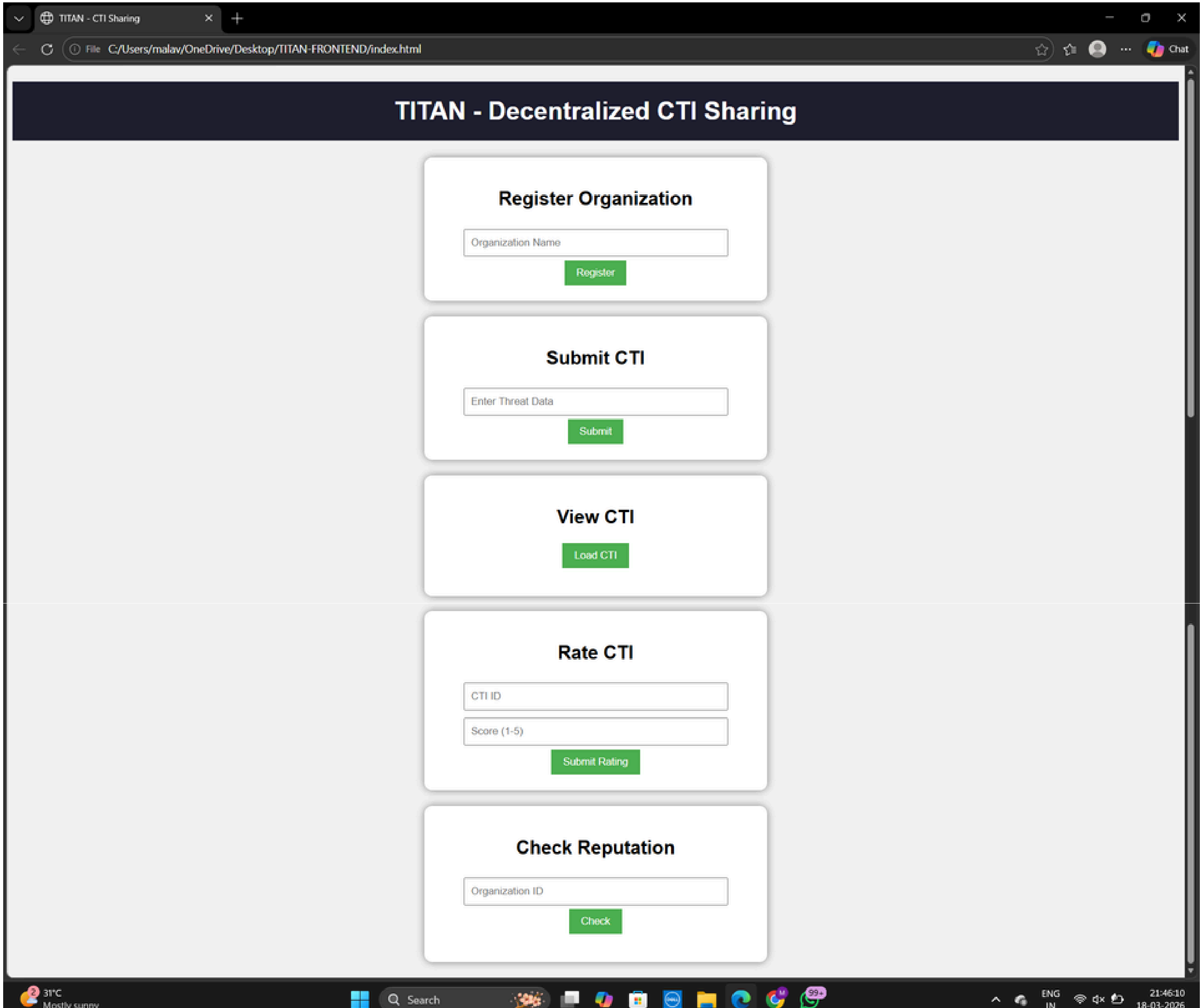
DATABASE / DATA DESIGN

ER DIAGRAM / SCHEMA



RESULTS & OUTPUT

Frontend



CONCLUSION

- **TITAN successfully bridges the gap between collaborative threat intelligence and strict zero-trust environments.**
- **By entirely removing centralized key management and databases, the system eliminates single points of failure.**
- **The integration of Ethereum smart contracts and the EigenTrust algorithm proves that automated, reputation-based CTI exchange is highly viable and prevents freeriding.**

FUTURE ENHANCEMENTS

- **Zero-Knowledge Proofs (ZKPs):**

  Implementing ZKPs to allow organizations to prove they possess valid threat intelligence without revealing their identity.

- **Layer 2 Deployment:**

  Migrating the smart contract to a Layer 2 network (like Polygon or Arbitrum) to drastically reduce transaction gas costs.

- **AI/ML Integration:**

  Deploying a machine learning model to pre-scan and validate threat submissions automatically before human peers rate them.

REFERENCES

- [1] Y. Wu, Y. Qiao, Y. Ye, and B. Lee, "Towards improved trust in threat intelligence sharing using blockchain and trusted computing," IOTSMS, 2019.
- [2] S. Li, T. Jiang, K. Chen, "Blockchain-based Secure Information Sharing Framework for Cyber Threat Intelligence.".
- [3] S. Kamvar, M. Schlosser, and H. Garcia-Molina, "The Eigen Trust algorithm for reputation management in P2P networks," WWW, 2003.
- [4] M. Patel, R. Shah, "Decentralized Cyber Threat Intelligence Sharing Using Smart Contracts.".
- [5] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," 2008.

THANK YOU